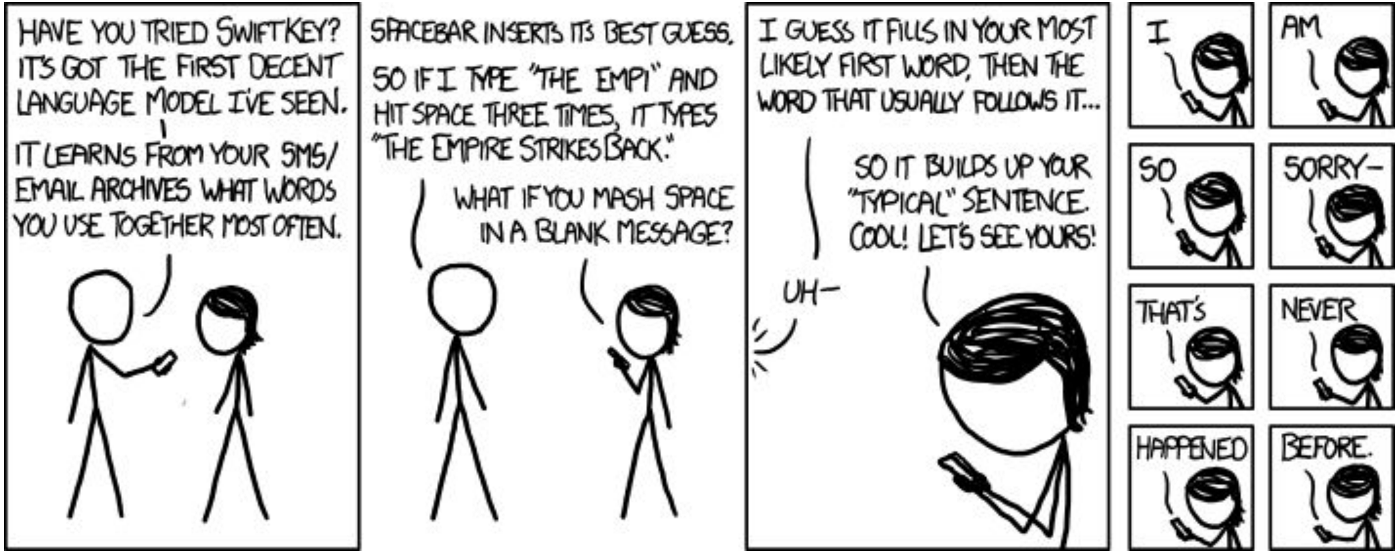
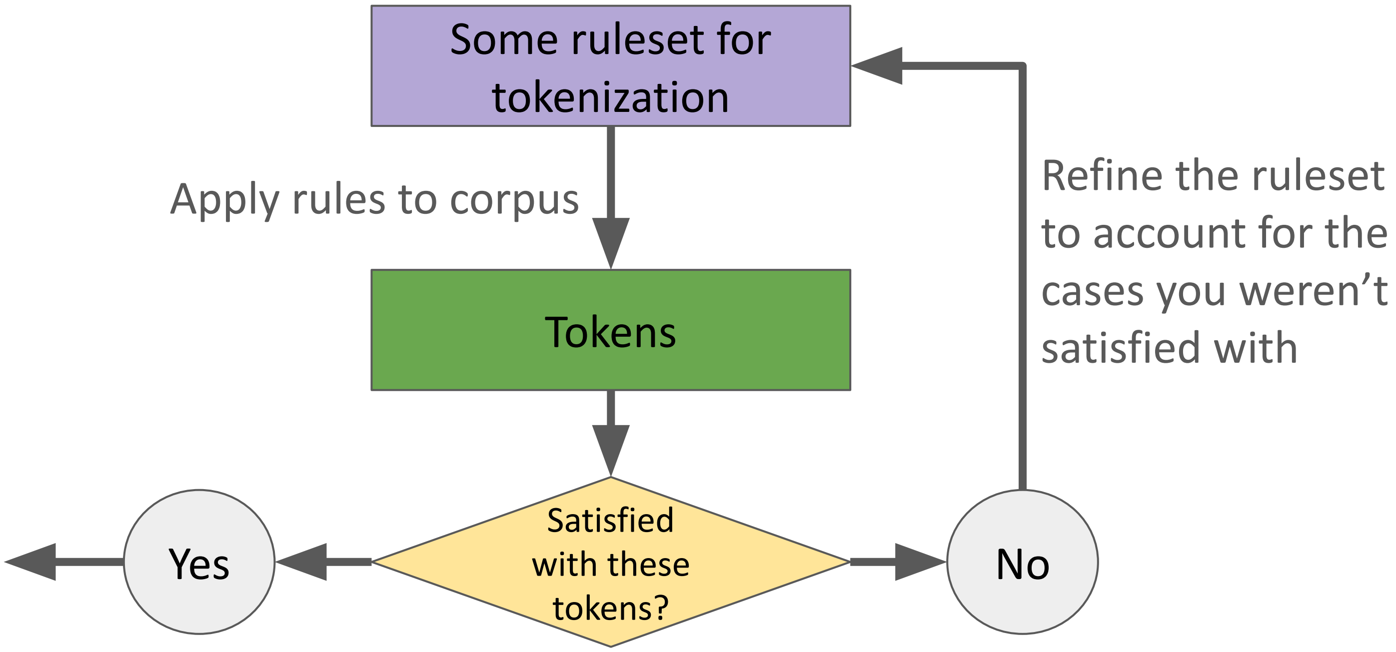
CS 159 Week 3:
Probability, N-Grams, and Smoothing

(https://xkcd.com/1068)



Where we left off



Done (for now!)
Keep this ruleset, and the set of resulting tokens as your vocabulary.

A journey from **words** to **language** to **meaning**

(we are here)

A journey from $\boxed{\textbf{words}}$ to **language** to **meaning**

Today’s agenda:

A journey from **words** to **language** to **meaning**

Attempt #1

Language consists of sequences of words.

Is this a real/valid English sentence?

"I teach at a university."

Is this a real/valid English sentence?

"Dsfy dshdyyore ncvgt aku asdgbvry."

Is this a real/valid English sentence?

"Berry berry berry berry berry berry berry."

Is this a real/valid English sentence?

"Provincialism regressive, caller abandon made."

Is this a real/valid English sentence?

“Colorless green ideas sleep furiously.”

(Example originally due to Noam Chomsky)

Attempt #1

Language consists of sequences of words.

Attempt #2

Language consists of **plausible** sequences of words.

Attempt #2

???

Language consists of **plausible** sequences of words.

More formalisms!

Given a sequence of k words (tokens):

$$w_1, w_2, w_3, \dots, w_k$$

To say this sequence is "plausible" is to say it has high probability:

$$P(w_1, w_2, w_3, \dots, w_k) >> 0$$

A missing ingredient

We need a mathematical model that lets us compute:

$$P(w_1, w_2, w_3, \ldots, w_k)$$

A missing ingredient

We need a mathematical model that lets us compute:

$$P(w_1, w_2, w_3, \ldots, w_k)$$

…We need a **language model (LM)**!


"QUICK! TO THE BAT- BOARD!
WATERSON

Word frequencies are actually useful!

- Word frequencies (and relative frequencies) aren't only used to create bad language models…
- They are also extremely useful tools for studying language!
- Some examples:
  - Authorship attribution (each individual author has biases towards certain words)
  - Spam detection (some words are more common in spam emails)
  - Trend analysis (ever seen a word cloud?)
- Fun visualization of word frequencies: Google Books Ngram Viewer

Improving our language model

`P(“Angeles”|”My”,”college”,…”Los”)`

“My college is located near Los Angeles.”

Improving our language model

P(“Angeles”|”My”,”college”,…”Los”)
≈
P(“Angeles”)

Independence assumption: none of this stuff matters

“~~My college is located near Los~~ Angeles.”

Improving our language model

P(“Angeles”|”My”,”college”,…”Los”)

Probably unimportant?

“My college is located near Los Angeles.”

Improving our language model

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Probably important!

Improving our language model

P("Angeles"|"My","college",…"Los")
≈
P("Angeles"|"Los")

"~~My college is located near~~ Los Angeles."

Markov assumption: only the previous event (previous word) matters

Improving our language model

P("Angeles"|"My","college",…"Los")

≈

P("Angeles"|"near","Los")

"~~My college is located~~ near Los Angeles."

Markov assumption: only the m previous events, for some constant m, matter

Improving our language model

P("Angeles"|"My","college",…"Los")
≈
P("Angeles"|$\underbrace{\text{"near","Los"}}_{\text{context}}$)

"~~My college is located~~ near Los Angeles."

Markov assumption: only the m previous events, for some constant m, matter

Improving our language model

$$P(\text{"Angeles"}|\text{"My","college",...,"Los"}) \approx P(\text{"Angeles"}|\underbrace{\text{"near","Los"}}_{\text{context}})$$

"~~My college is located~~ near Los Angeles."

Markov assumption: only the m previous events, for some constant m, matter

For this to work, we need to count *sequences* of words…

From last week…

Individual characters	e.g., “r”
Sequences of characters (for some languages, these are known as *subwords*)	e.g., “大学”
Words (as defined by some vocabulary)	e.g., “strawberry”
Sequences of words (aka *n-grams*)	e.g., (“strawberry”, “jam”), representing “strawberry jam”

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aha!


"QUICK! TO THE BAT- BOARD!

More context, more better?

- Bigram models are what you all may remember building in CS5!
- From your CS5 experience, you can hopefully believe the bigram model generates more realistic text than the bag-of-words model
- Does this scale? Can we get even better text from a trigram model? A 4-gram model? A 42-gram model?
- Theoretically yes, but…

Tragic news: data isn't infinite

$$P(\text{“Angeles”} \mid \text{”My”},\text{”college”},\text{”is”},\text{”located”},\text{”near”},\text{”Los”})$$

Tragic news: data isn't infinite

$$P(\text{“Angeles”} \mid \text{”My”}, \text{”college”}, \text{”is”}, \text{”located”}, \text{”near”}, \text{”Los”})$$

$\big\downarrow$ (need to compute)

$$C(\text{”My”}, \text{”college”}, \text{”is”}, \text{”located”}, \text{”near”}, \text{”Los”}, \text{“Angeles”})$$

Tragic news: data isn't infinite

P("Angeles"|"My","college","is","located","near","Los")

(need to compute)

C("My","college","is","located","near","Los","Angeles")

How many times would you see this exact sequence in the data?

Tragic news: data isn't infinite

P("Angeles"|"Seven","cat","cafes","are","located","near","Los")

(need to compute)

C("Seven","cat","cafes","are","located","near","Los","Angeles")

How many times would you see this exact sequence in the data?

Tragic news: data isn't infinite

`C(”Seven”,”cat”,”cafes”,”are”,”located”,”near”,”Los”,“Angeles”)`

How many times would you see this exact sequence in the data?

- This is a fundamental problem of data sparsity
- In this week's readings, you'll see one naive solution: *smoothing*
- But in general, arguably the whole mission of modern NLP is making models robust to sparse contexts!

Where we’re headed

- End goal: modern language models (aka **Large** Language Models, or LLMs)
- Weeks 4 and 5: Interesting things you can do using only basic word frequencies (Markov and even bag-of-words models)
- Weeks 6-8: Some other neat NLP topics *seemingly* unrelated to language models

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- Week 9ish: